



Gauss' Law

Prepared by:

Dr. Yasser El-Batawy

Electric Flux for a Closed Surface

- The net electric flux through any *closed surface* is equal to the net charge enclosed by the closed surface divided by the permittivity of the free space.
- Then, the Gauss' Law can be written as

$$\varphi_E = \frac{\sum Q_{en}}{\epsilon_0}$$

$$\varphi_E = \oint \underline{E} \cdot \underline{dA} = \frac{\sum Q_{en}}{\epsilon_0}$$

Electric Flux for a Closed Surface

- Although Gauss's law can, in theory, be solved to find $\mathbf{E}$ for any charge configuration, in practice it is limited to symmetric situations

$$\varphi_E = \oint \underline{E} \cdot \underline{dA} = \frac{\sum Q_{en}}{\epsilon_0}$$

How to get E Using Gauss' Law

- Determine the direction of E from Symmetry.
- Sketch a Gaussian closed surface that suits the symmetry of the problem.
- The surface should be chosen such that the angle between the field and the various parts of the surface would be either zero or ninety degrees, and the magnitude of the field is constant

$$\oint \underline{E} \cdot \underline{dA} = \frac{\sum Q_{en}}{\epsilon_0}$$

How to get E Using Gauss' Law

- Calculate the enclosed charge by the Gauss' surface
- Apply Gauss' law on the closed surface.

$$\oint \underline{E} \cdot \underline{dA} = \frac{\sum Q_{en}}{\epsilon_0}$$

Example 1:

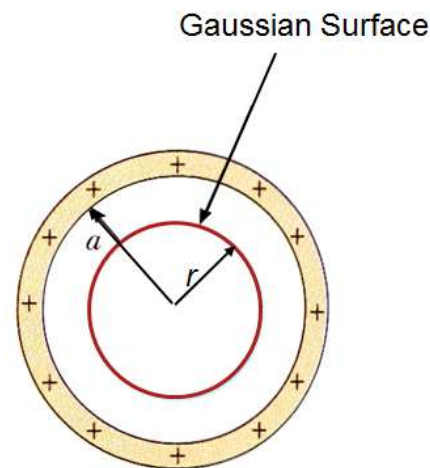
- A spherical surface

A thin spherical surface of radius r_0 possesses a total net charge Q that is uniformly distributed on it. Determine the electric field at points

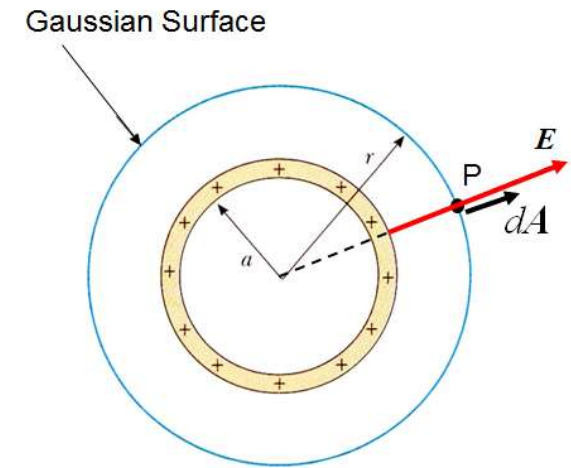
- (a) outside the sphere
- (b) Inside the sphere

Solution

- To find the electric field at any point P we draw a suitable Gaussian surface passing through the point.
- In the present case the surface that suits the symmetry is a spherical one.
- From symmetry one can see that the electric field at any point on a spherical Gaussian surface is along the radius of the sphere



The field point is inside the shell



The field point is outside the shell

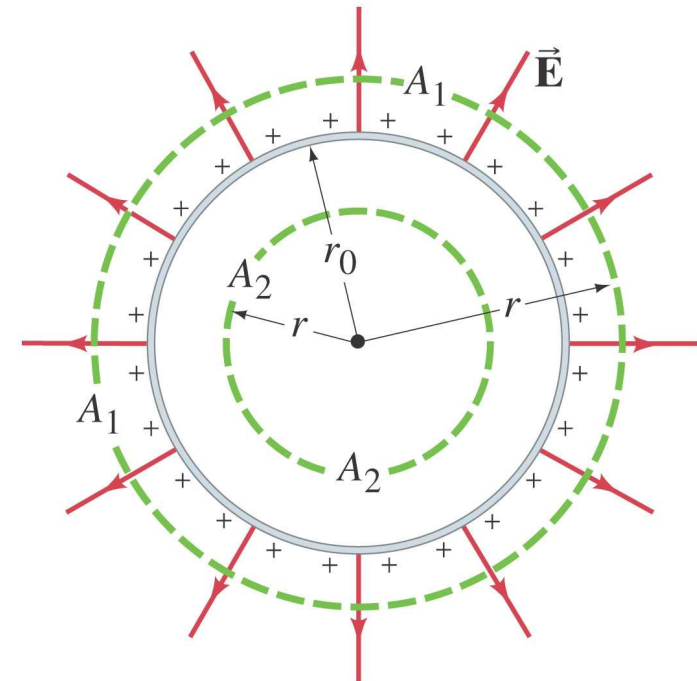
Solution

- The surfaces A_1 and A_2 represent the two gaussian surfaces we use to determine $\mathbf{E}$ for $(r > r_o)$ and $(r < r_o)$, respectively

- For $r < r_o$

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{in}}{\epsilon_o} = 0$$

$$\mathbf{E} = 0$$



Solution

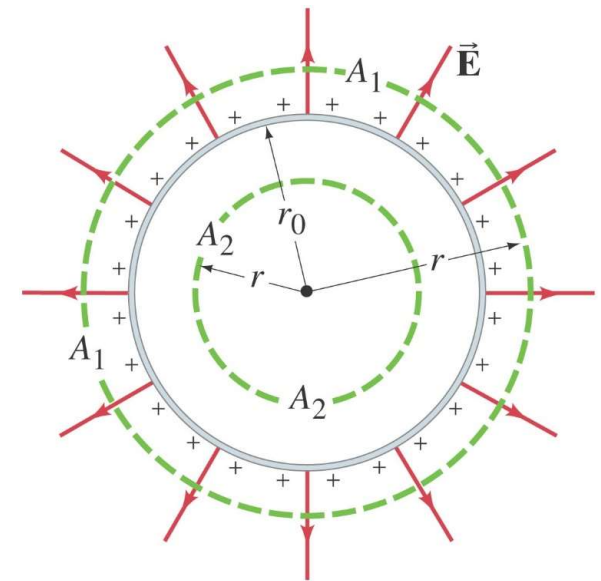
○ For $r > r_o$

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = \oint E \cos 0^\circ dA = \oint E dA = \frac{q_{in}}{\epsilon_o}$$

From symmetry the magnitude of the field is the same for all points on the spherical surface

$$\therefore \Phi_E = E \oint dA = E 4\pi r^2 = \frac{Q}{\epsilon_o}$$

$$\therefore E = \frac{Q}{4\pi\epsilon_o r^2}$$

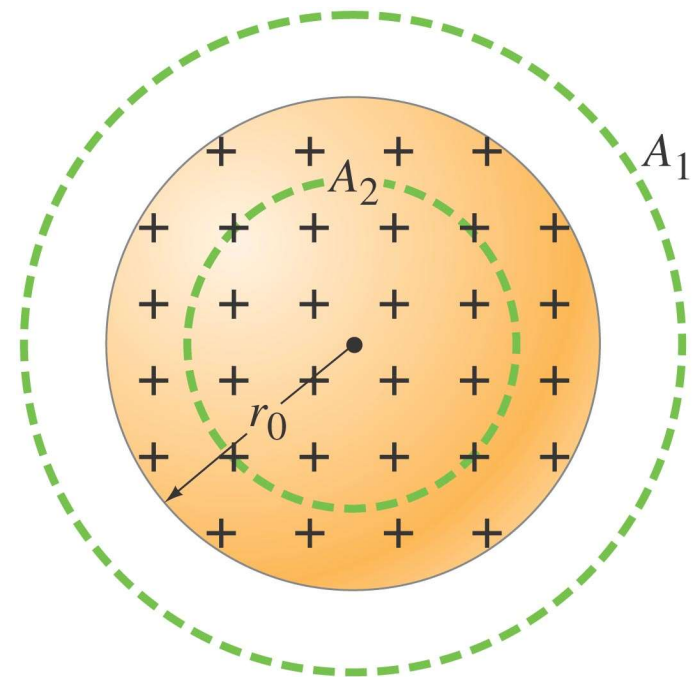


Example 2 :

• Solid sphere of charge.

A spherical volume of radius r_0 is charged uniformly with ρ C/m³. Determine the electric field

- (a) inside the sphere ($r < r_0$)
- (b) outside the sphere ($r > r_0$).



Solution

For $r < r_o$

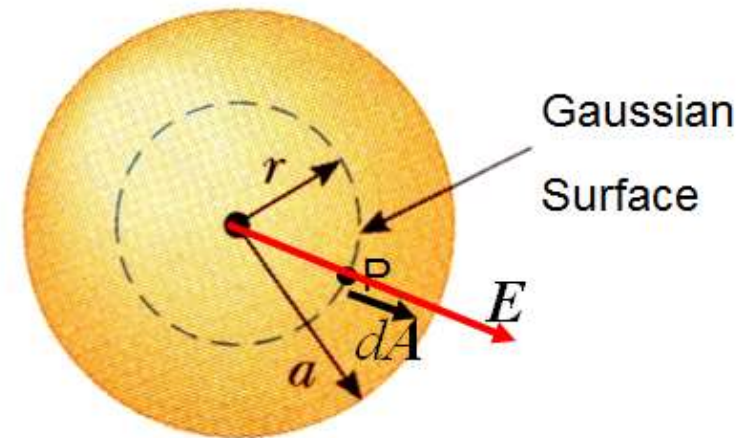
$$\therefore \Phi_E = E \oint dA = E(4\pi r^2)$$

$$\therefore \Phi_E = \frac{q_{in}}{\epsilon_o} = \frac{(4\pi r^3 / 3)\rho}{\epsilon_o} \quad \text{Gauss' law}$$

$$\therefore E(4\pi r^2) = \frac{4\pi r^3 \rho}{3\epsilon_o}$$

$$\therefore E = \frac{\rho r}{(3\epsilon_o)}$$

$$\therefore \mathbf{E} = \frac{\rho r}{3\epsilon_o} \hat{r}$$



A field point inside the charged volume

Solution

○ For $r > r_o$

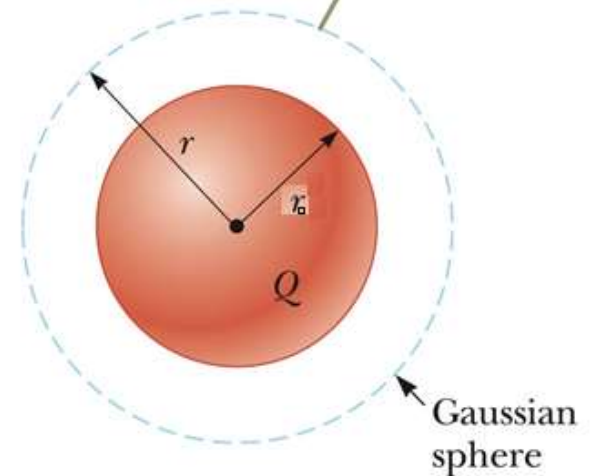
$$\Phi_E = E \oint dA = E(4\pi r^2) = \frac{q_{in}}{\epsilon_o}$$

$$q_{in} = \rho \frac{4\pi}{3} r_o^3$$

$$\therefore E(4\pi r^2) = \rho \frac{4\pi}{3\epsilon_o} r_o^3$$

$$\mathbf{E}_{out} = \frac{\rho r_o^3}{3\epsilon_o r^2} \hat{\mathbf{r}}$$

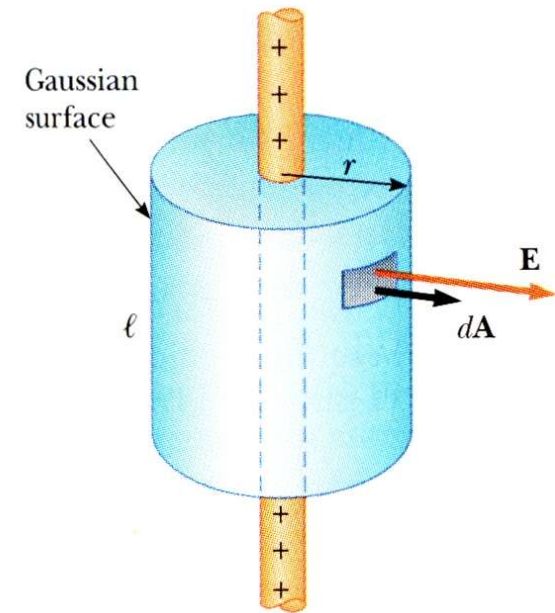
For points outside the sphere, a large, spherical gaussian surface is drawn concentric with the sphere.



Example 3:

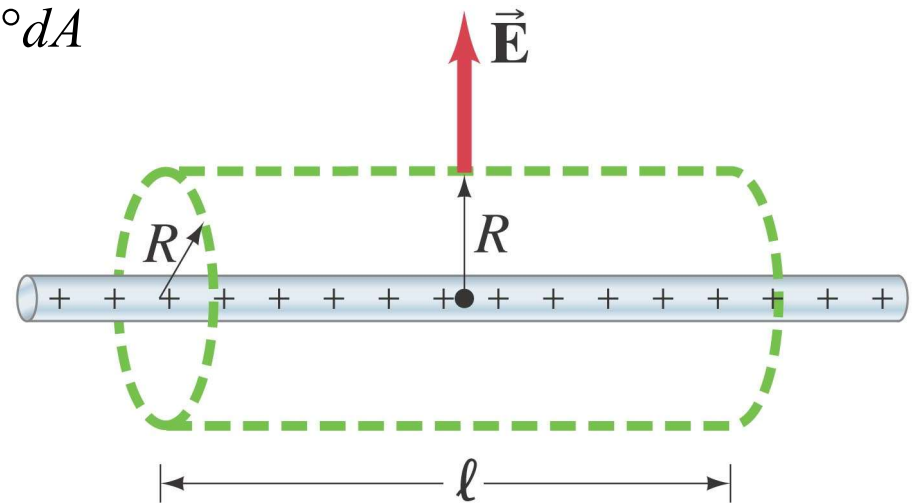
◦ Long uniform line of charge.

A very long straight wire possesses a uniform positive charge per unit length, λ . Calculate the electric field at points near the wire.



Solution

$$\begin{aligned}
 \Phi_E &= \oiint E \cdot d\mathbf{A} \\
 &= \int_{\text{upper base}} E \cdot d\mathbf{A} + \int_{\text{lower base}} E \cdot d\mathbf{A} + \int_{\text{cylindrical surface}} E \cdot d\mathbf{A} \\
 \Phi_E &= \int_{\text{upper base}} E \cos 90^\circ dA + \int_{\text{lower base}} E \cos 90^\circ dA \\
 &\quad + \int_{\text{cylindrical surface}} E \cos 0^\circ dA \\
 \Phi_E &= 0 + 0 + \int_{\text{cylindrical surface}} E dA
 \end{aligned}$$



Solution

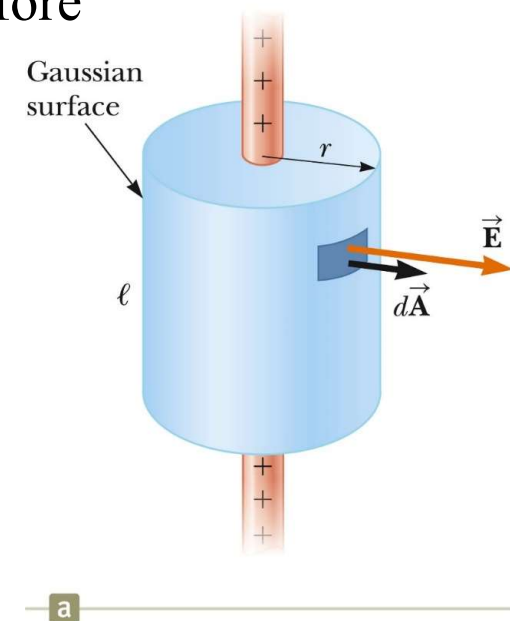
From symmetry it is clear that the magnitude of the field is the same for all points on the cylindrical surface. Therefore

$$\Phi_E = E \int_{\text{cylindrical surface}} dA = E(2\pi r l)$$

Applying Gauss' law, $\Phi_E = \frac{q_{in}}{\epsilon_o}$

$$\therefore E(2\pi r l) = \frac{\lambda l}{\epsilon_o}$$

$$E = \frac{\lambda}{2\pi\epsilon_o r} \hat{r}$$

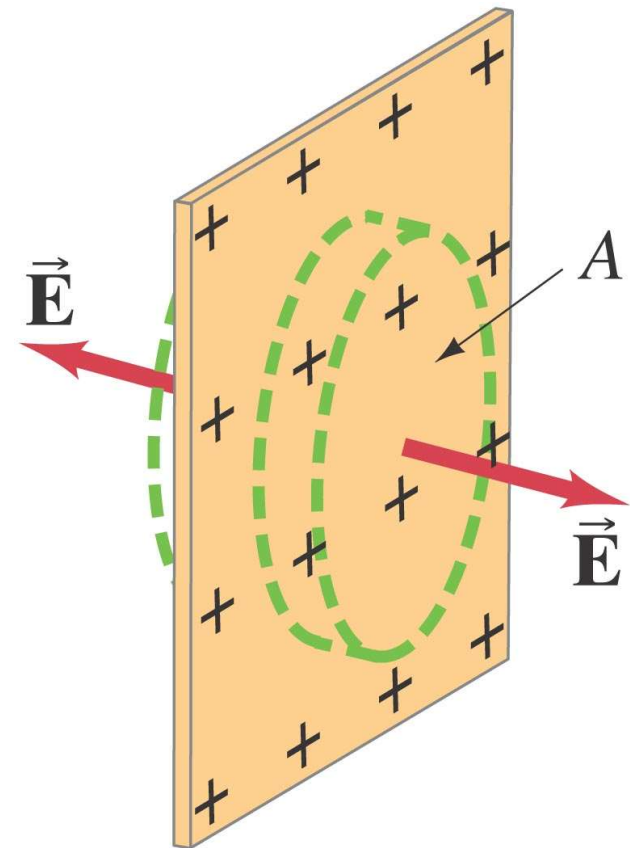


Example 4:

- **Infinite plane of charge.**

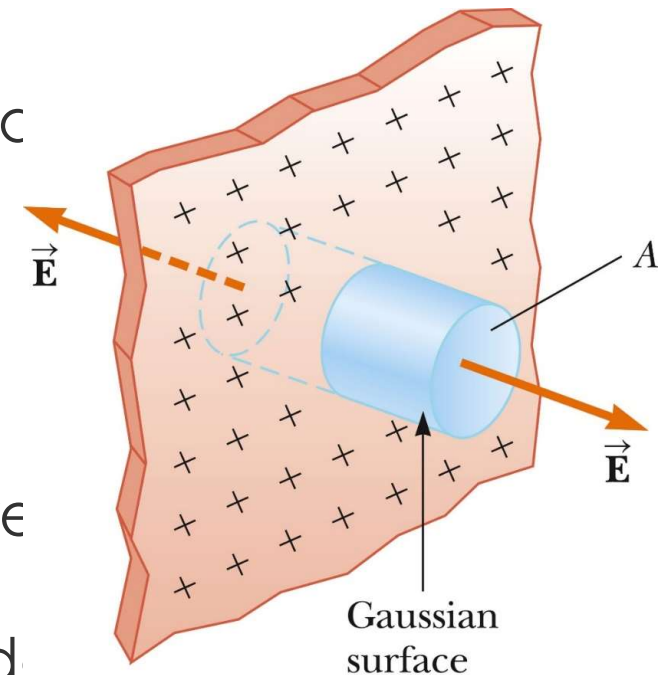
Charge is distributed uniformly, with a surface charge density σ (σ = charge per unit area) over a very large but very thin nonconducting flat plane surface.

Determine the electric field at points near the plane.



Solution

- $\vec{E}$ must be perpendicular to the plane and must have the same magnitude at points equidistant from the plane.
- Choose a small cylinder whose axis is perpendicular to the plane for the Gaussian surface.
- $\vec{E}$ is parallel to the curved surface and there is no contribution to the surface area from this curved part of the cylinder.
- The flux through each end of the cylinder is EA and so the total flux is $2EA$.



Solution

$$\Phi_E = \oiint E \cdot d\mathbf{A}$$

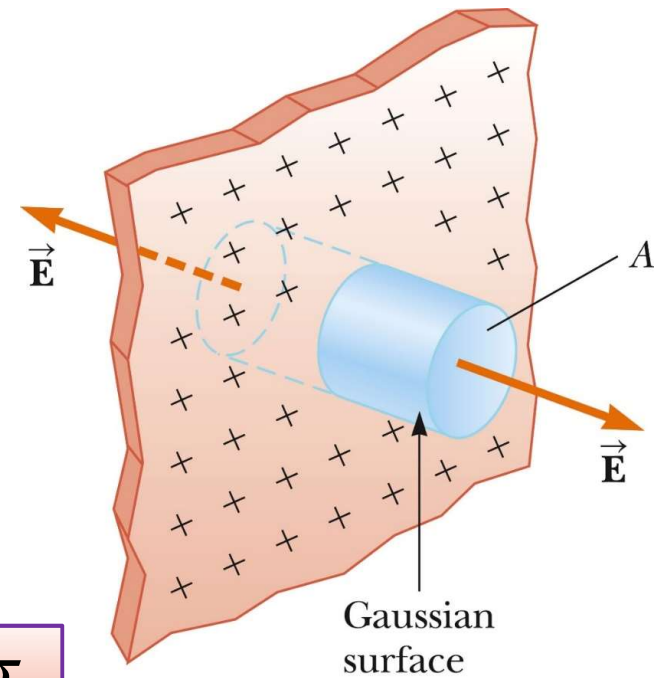
$$\Phi_E = \int_{\text{front base}} E \cdot d\mathbf{A} + \int_{\text{back base}} E \cdot d\mathbf{A} + \int_{\text{cylindrical surface}} E \cdot d\mathbf{A}$$

$$\Phi_E = \int_{\text{front base}} E \cos 0^\circ dA + \int_{\text{back base}} E \cos 0^\circ dA + \int_{\text{cylindrical surface}} E \cos 90^\circ dA$$

$$\therefore \Phi_E = E \int_{\text{front base}} dA + E \int_{\text{back base}} dA = 2EA$$

$$\therefore \Phi_E = \frac{q_{\text{in}}}{\epsilon_0} \therefore 2EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$



Check

<https://www.youtube.com/watch?v=ln3E9VfxhJA>

<https://www.youtube.com/watch?v=9J3Wh8YRYWo>